FAKE NEWS
DETECTION SYSTEM

Using Machine Learning & NLP

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Python | Flask | Scikit-Learn | NLP

Introduction & Problem Statement

Fake news spreads rapidly through social media platforms, creating confusion and misinformation. Verifying every news article manually is time-consuming and impractical — making an automated, intelligent detection system the need of the hour.

Rapid Spread

Fake news travels 6x faster than real news on social media platforms

Manual Slowness

Human fact-checkers cannot keep pace with hundreds of daily stories

Automation Gap

No simple, accessible tool exists for common users to verify news

This project solves the problem using Python · Flask · Machine Learning · NLP

Project Objectives

Auto Detection

Automatically identify fake and real news without human intervention

Reduce Misinformation

Help users avoid believing and sharing incorrect information

Fast & Accurate

Provide near-instant predictions with high accuracy rates

Web Application

Build a user-friendly Flask-based interface accessible to all

Technologies & Tools Used

Python

Core Language

Flask

Web Framework

Pandas

Data Handling

NumPy

Numerical Ops

Scikit-Learn

ML Models

NLTK

NLP Processing

HTML/CSS

Frontend

Bootstrap

UI Design

Pickle

Model Saving

System Architecture / Workflow

1

User Input

News article text entered by user

2

Text Preprocessing

Lowercase · Remove punctuation · Stopwords

3

TF-IDF Extraction

Text converted to numerical feature vectors

4

ML Model

Logistic Regression classifies the article

5

Prediction Result

✓Real News or ✗Fake News

Dataset & Data Preprocessing

Dataset Information

Source	Fake & Real News CSV Files
Size	Thousands of labelled records
Format	CSV (Title · Text · Label)
Labels	FAKE / REAL
Split	80% Training · 20% Testing

Preprocessing Steps

1. **Lowercase**
   Convert all text to lowercase for uniformity
2. **Remove Punct.**
   Strip punctuation & special characters
3. **Remove Stopwords**
   Drop common words (the, is, at...)
4. **Tokenization**
   Split text into individual words/tokens
5. **Stemming**
   Reduce words to their root form

Feature Extraction & ML Model

TF-IDF Vectorization

What is TF-IDF?

TF-IDF (Term Frequency–Inverse Document Frequency) converts raw text into numerical feature vectors that Machine Learning models can understand.

- Captures word importance across documents
- Reduces noise from common words
- Lightweight and fast to compute
- Improves classification accuracy

Machine Learning Models

Logistic Regression

✓Selected — Best Accuracy

Naive Bayes

Fast but lower accuracy

Passive Aggressive

Good for streaming data

Model saved as .pkl file using Pickle library

Model Training & Flask Web Application

Training Process

Step 1 — **Load Dataset**
Import Fake & Real CSV files using Pandas

Step 2 — **Preprocess**
Clean, tokenize and normalize the text

Step 3 — **TF-IDF**
Fit vectorizer and transform features

Step 4 — **Train Model**
Fit Logistic Regression on training data

Step 5 — **Save Model**
Serialize using Pickle for web app reuse

Flask Web Application

Clean UI
Minimal, responsive news input form

Instant Results
Prediction within milliseconds

Responsive
Mobile-friendly Bootstrap design

Reliable
Pre-loaded model — no training delay

Accessible
Runs on localhost or cloud server

Working Demo & Results

Fake News Detector — Web Interface

Enter News Article Here...

"Breaking: Scientists confirm major discovery that challenges existing theories about climate change. Independent verification pending from multiple research institutes..."

Predict

✓ **REAL NEWS**

✗ **FAKE NEWS**

Model Performance

90–98%

Classification Accuracy

High Precision	**High** Recall
0.93+ F1 Score	**< 1 sec** Response Time

Advantages, Limitations & Future Scope

✓ Advantages

- Fast automated detection
- Easy-to-use web interface
- Reduces spread of misinformation
- Scalable to large datasets
- No technical knowledge needed

⚠️ Limitations

- Accuracy tied to data quality
- New fake patterns undetected
- Requires periodic retraining
- English-only at present
- Context/satire may be misread

🚀 Future Scope

- BERT / Transformer models
- Social media API integration
- Multi-language support
- Real-time fact checking
- Browser extension plugin

Conclusion

Successfully built:

A complete Fake News Detection system using ML and NLP

Core Technology:

Logistic Regression on TF-IDF features — 90-98% accuracy

Web App:

Flask-based interface for real-time user predictions

Impact:

Reduces misinformation; scalable and easy to use

Thank You!

Dhananjay Sharma

Fake News Detection System | MCA Final Year Project

Questions & Feedback Welcome

- Python
- Flask
- Scikit-Learn
- NLTK
- TF-IDF